



Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads

Problem statement ID: 26037

Team name- Epsilon



Problem statement

Design and simulate an adaptive path planning system for an autonomous vehicle that operates in unstructured Indian road conditions. The system should perceive the environment using a multi-sensor setup such as camera, LiDAR, and radar, and identify diverse road users and obstacles, including auto-rickshaws, pushcarts, pedestrians, and animals. It should predict the short-term motion of surrounding agents, including non lane-based and irregular movement patterns, and generate a safe, collision-free path that can be replanned in real time. The solution should also handle practical driving situations such as missing lane markings, informal merging, sudden pedestrian movement, and unexpected obstacles on the road. Teams should validate their solution using at least five realistic Indian road scenarios: an unmarked village road, a busy urban intersection without signals, a highway merge involving slow-moving vehicles, a dense market area with mixed traffic, and a sudden cattle-crossing event. Teams are encouraged to use MathWorks tools such as RoadRunner for scenario design, Automated Driving Toolbox for sensor modeling and fusion, Navigation Toolbox and Stateflow for planning and decision logic, Vehicle Dynamics Blockset or a Simulink bicycle model for vehicle behavior, and Deep Learning Toolbox for detection and trajectory prediction.

Perception: Challenges Faced in Indian Roads



Objective: Overcome the collapse of standard lane priors and visual occlusion by fusing 2D semantic classification with 3D physical depth mapping to track heterogeneous traffic and identify structural road hazards.

- **The Collapse of Structural Priors:** Modern autonomous perception systems rely mathematically on white lane dividers, raised concrete curbs, and predictable intersection geometries. On Indian roads, these priors instantly fail. The system must transition from "lane-based perception" to "semantic free-space perception," as road edges are frequently unclear, unpaved, or occupied by static obstacles, rendering standard high-definition mapping algorithms useless.
- **Taxonomy & Extreme Heterogeneity:** Western datasets use rigid taxonomies (Car, Pedestrian, Cyclist). Indian traffic introduces auto-rickshaws, custom pushcarts, slow tractors, and wandering livestock. Without explicit training, standard neural networks ignore or dangerously misclassify these agents. The vision backbone must be fine-tuned specifically on the **Indian Driving Dataset (IDD)** to establish a correct, localized visual taxonomy.
- **Severe Occlusion & Track Fragmentation:** Indian driving behavior prioritizes maximum spatial utilization and tight gap-filling, creating extreme, continuous occlusion. When a camera only sees the roof of an auto-rickshaw hidden behind a bus, classical trackers drop the bounding box, causing "ID switching." Overlap-based tracking architectures (**ByteTrack**) paired with **Interacting Multiple Model (IMM)** filters are required to maintain persistent mathematical estimates of occluded vehicles based on their last known kinematics.
- **The 3D Geometry Problem (Potholes vs. Shadows):** A standard 2D camera suffers from depth ambiguity; it cannot mathematically differentiate a harmless tree shadow from a highly destructive, deep pothole. To calculate physical drivability, the perception system must utilize 3D point-cloud data (via **PointPillars**) to measure the negative Z-axis geometry of the road surface, translating dark pixels into quantifiable "Static High-Cost Tokens" (negative obstacles).
- **The Architectural Imperative:** The perception layer must decouple from lane priors. It requires the integration of PointPillars for dense 3D surface mapping, IDD-trained YOLOv8 detectors for 2D semantic classification, and IMM tracking filters within Simulink to output highly stable, 3D agent track queries to the downstream planner.

Proposed Solution for Perception



Objective: Transform unstructured sensor data into a unified 3D Bird's-Eye View (BEV) to accurately track heterogeneous Indian traffic, define unmarked road boundaries, and measure surface hazards.

- **PointPillars & YOLO Integration (3D Agent Detection):** The Automated Driving Toolbox synthesizes raw LiDAR and camera data from the RoadRunner environment. The LiDAR point cloud is processed by a PointPillars architecture into a dense 2D pseudo-image to predict 3D bounding boxes. Concurrently, a YOLOv8 network—fine-tuned on the Indian Driving Dataset (IDD)—processes the RGB feed to classify uniquely Indian traffic such as auto-rickshaws, pushcarts, and cattle.
- **Bipolar Elevation Profiling (Potholes & Speed Breakers):** Instead of standard 2D segmentation, the PointPillars BEV grid analyzes vertical variance (Δz) in the LiDAR data. The system classifies severe negative spikes ($\Delta z < 0$) as potholes and positive spikes ($\Delta z > 0$) as unmarked speed breakers or road debris. These are extracted as physical tokens, allowing the planner to calculate if the ego-vehicle can safely straddle or clear the hazard.
- **Semantic Free-Space Estimation (Unmarked Roads):** To handle rural village roads without lane markings, LiDAR ground-plane fitting and camera segmentation are fused to create a continuous polygon of drivable tarmac. This defines the physical corridor boundaries (e.g., dirt shoulders, open drains) for the downstream costmap.
- **3D ByteTrack, Radar Doppler & IMM State Estimation:** To manage severe traffic occlusion and erratic swerving, 3D bounding boxes are associated across frames using ByteTrack. These tracks are fed into an Interacting Multiple Model (IMM) filter via the Sensor Fusion and Tracking Toolbox. By directly integrating instantaneous Radar Doppler velocity (v_r), the IMM filter eliminates velocity-estimation lag during sudden two-wheeler cut-ins, dynamically switching between kinematic models (constant velocity vs. turn-rate) to maintain a stable track lock.
- **Core Literature:**
 - Varma et al. (2019). "IDD: A Dataset for Exploring Problems of Autonomous Navigation in Unconstrained Environments".
 - Lang et al. (2019). "PointPillars: Fast Encoders for Object Detection from Point Clouds".
 - Zhang et al. (2022). "ByteTrack: Multi-Object Tracking by Associating Every Detection Box".

The Trajectory Prediction Crisis & Non-Lane Motion Modeling



Objective: Replace deterministic, lane-anchored motion models with multi-modal spatial probability networks capable of anticipating the opportunistic gap-filling and hazard-induced swerving inherent to mixed traffic.

- **The Failure of Lane-Centric Kinematics:** Standard trajectory prediction models act like trains on invisible tracks, anchoring their math to lane centerlines. Because formal lane discipline is virtually non-existent in Indian traffic, these algorithms fail catastrophically. The system must predict motion in an unconstrained, two-dimensional coordinate space where any physical tarmac is a valid path.
- **Micro-Gap Filling & High-Frequency Lateral Dynamics:** Indian traffic features opportunistic "gap-filling," where two-wheelers and auto-rickshaws execute high-frequency lateral swerves to exploit micro-gaps in moving traffic. These agents exhibit extreme non-holonomic agility that breaks standard Kalman filter prediction envelopes, requiring sequence models trained explicitly on real-world Indian swerving behaviors.
- **Causal Opacity & Hazard-Induced Swerving:** To a standard AI, a vehicle violently swerving on an empty road looks like unlearnable noise. The hidden cause is often a deep pothole or an unpaved shoulder. If the perception system does not explicitly feed these "static hazards" into the prediction network, the AI cannot learn the causal relationship. The architecture must force cross-attention between the moving agents (Track Queries) and road defects (Map Queries) to accurately predict evasive maneuvers.
- **Multi-Modal Uncertainty (The "Freezing Robot" Antidote):** Indian traffic relies on informal negotiation (e.g., nudging forward at a junction to force right-of-way) rather than rigid traffic signals, making the exact future of an agent mathematically unknowable. If an AI attempts to predict a single deterministic line, it will frequently be wrong and abruptly brake, suffering from "Freezing Robot Syndrome."

Proposed Solution: Prediction/Trajectory

Objective: Predict the multi-modal, non-lane-based future paths of surrounding agents using self-attention mechanisms in 3D space.

- **Attention in BEV Space (The MotionFormer):** Emulating the UniAD architecture, this pipeline utilizes a "MotionFormer"—a specialized Transformer module that acts as an interaction engine by launching learnable agent queries to cross-reference other moving vehicles and the static road topology. By executing this cross-attention directly in the 3D BEV coordinate space (via MATLAB's `selfAttentionLayer`), the network evaluates the spatial relationship between the IMM Track Queries (vehicles) and the 3D Map Queries (potholes and road boundaries).
- **Causal Hazard Avoidance:** The Transformer is fine-tuned exclusively on the Mendeley Chennai mixed-traffic trajectory dataset. By forcing the network to attend to the static Pothole Tokens, it mathematically learns the causal dynamics of unstructured roads: *If a two-wheeler approaches a negative-Z pothole, the probability of a lateral X-Y swerve increases.*
- **Multi-Modal Future Generation (GMM):** Because Indian traffic is highly non-deterministic, the prediction module outputs a Gaussian Mixture Model (GMM) representing spatial probability rather than a single fixed path.
 - **The Math:** For each agent, it predicts K possible future trajectories over a set time horizon (e.g., t+3seconds). Each trajectory mode outputs a sequence of (x,y) coordinates, a spatial covariance matrix (representing a physical uncertainty ellipse around the vehicle at each timestep), and an overall probability weight summing to 1.
 - **Costmap Integration:** These GMM spatial probability ellipses are projected directly into the downstream planner's `vehicleCostmap`. This allows the ego-vehicle to mathematically calculate exactly how much "safe clearance" it needs to leave around an erratic auto-rickshaw based on the neural network's uncertainty bounds.
- **Core Literature:**
 - *Hu et al. (2023).* "Planning-oriented Autonomous Driving" (UniAD).
 - *Canagaraj et al. (2015).* "Trajectory Data and Flow Characteristics of Mixed Traffic".

Multi-Modal Tracking & Occlusion Management

Unified State Estimation & Kinematic Resilience in High-Density Indian Traffic

1. Asynchronous Multi-Sensor Fusion

Stream Alignment & Association

- **Sensor Sync:** Harmonizes Camera (~30 Hz), Radar (~20 Hz), and LiDAR (~10 Hz) into [ISO 8855 frame](#) (X fwd, Y left, Z up).
- **Inter-Frame Propagation:** Time-stamped state propagation projects hits forward to arrival instant prior to measurement update.

Two-Stage ByteTrack Association

- **Stage 1 (High-Confidence):** Matched via Hungarian algorithm with Mahalanobis distance gating ($S = HPH^T + R$).
- **Stage 2 (Low-Confidence):** Leftover tracks associated with lower-confidence detections to prevent tracking loss in dust, shadows & clutter.

[Zhang et al., "ByteTrack: Multi-Object Tracking by Associating Every Detection Box."](#)

2. Maneuver Estimation (4-Model IMM)

Parallel Estimator Suite

- **CV:** Constant Velocity for smooth, steady driving.
- **CTRV:** Coordinated Turn Rate & Velocity for sharp cuts & swerves.
- **CA:** Constant Accel for aggressive braking/acceleration.
- **Stop Model:** Stationary traffic, roadside vehicles & cattle.

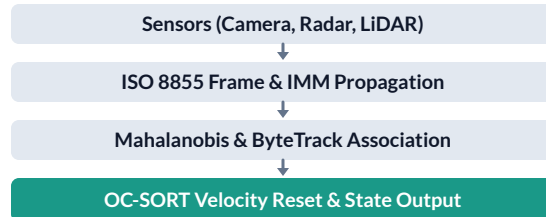
Occlusion Handling (OC-SORT)

- **Bounded Coasting:** Occluded tracks coast ≤ 10 frames (~1.0s), preventing infinite covariance expansion.
- **Virtual Trajectory Snapping:** Uses displacement vector between pre-loss & re-observation to fix velocity drift.
- **Dynamic Transition:** Updates model probabilities without resetting tracker filter states.

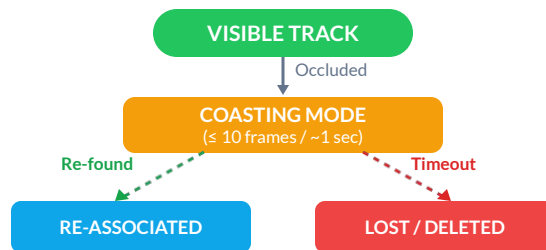
[Cao et al., "Observation-Centric SORT: Rethinking SORT for Robust Multi-Object Tracking."](#)

3. Pipeline Flow & State Transition

System Architecture Flow



Track Status State Machine



3-Stage Multimodal Pothole Detection & Costmap Injection

Cost-Effective Geometric Cavity Extraction & Trajectory-Aware Safety Arbitration

1. Perception Pipeline

Negative Obstacle Challenge:

Potholes dip downward ($1/R^2$ visibility drop); low mounting hides far rim.

Stage 1 – 2D Proposal & Far-Field Warning:

YOLOv8-seg performs real-time segmentation. [ASAHI Camera Slicing \(~4–6 Hz\)](#): Dynamically tiles high-res frames into 6–12 adaptive patches to detect candidate depressions at 15–30 m range as low-confidence caution regions prior to LiDAR verification.

Stage 2 – Surface & Curvature (LiDAR):

- **PatchWork++**: Concentric circular patches handle slope/bumps.
- **KD-Tree Curvature**: $C = \lambda_{\min} / \Sigma \lambda_i > 0.02$ & depth > 3 cm isolates cavity bowls.

		Accuracy (%)	s
Manual measurement	460	N/A	
High resolution	460	100.00	
10% reduction	415	100.00	
25% reduction	371	100.00	
33% reduction	301	100.00	
50% reduction	205	100.00	
75% reduction	109	73.00	

Detection performance of detected potholes

2. Costmap & Resilience

Graduated Costmap Injection:

- **Shallow (< 5 cm)**: Soft-cost (0.1–0.4) triggers longitudinal slowdown without swerving.
- **Deep (> 5 cm)**: High-cost (0.8–1.0) inflates impassable walls for Hybrid A* path avoidance.

Safety Arbitration (Indian Context):

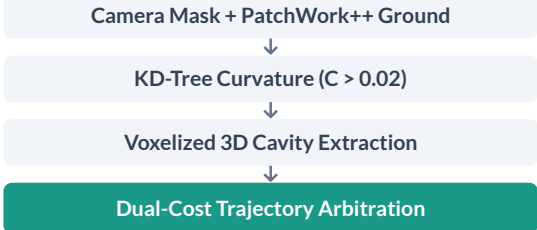
Pedestrian/cattle risk costs strictly override swerving to prevent collisions.

Indirect Perception & Fallbacks:

- **Lead Vehicle Mining**: Analyzes pitch drops & swerves of leading cars to infer occluded holes.
- **IMU Dead-Reckoning**: Maintains pose continuity during 2–5s GNSS outages.

3. Pipeline & Validation

Pipeline Flow Schematic:



Feasibility & Benchmark:

Faisal & Gargoum (2025) Proof on sparse automotive LiDAR point clouds.

100%

DETECTION

0.6 mm

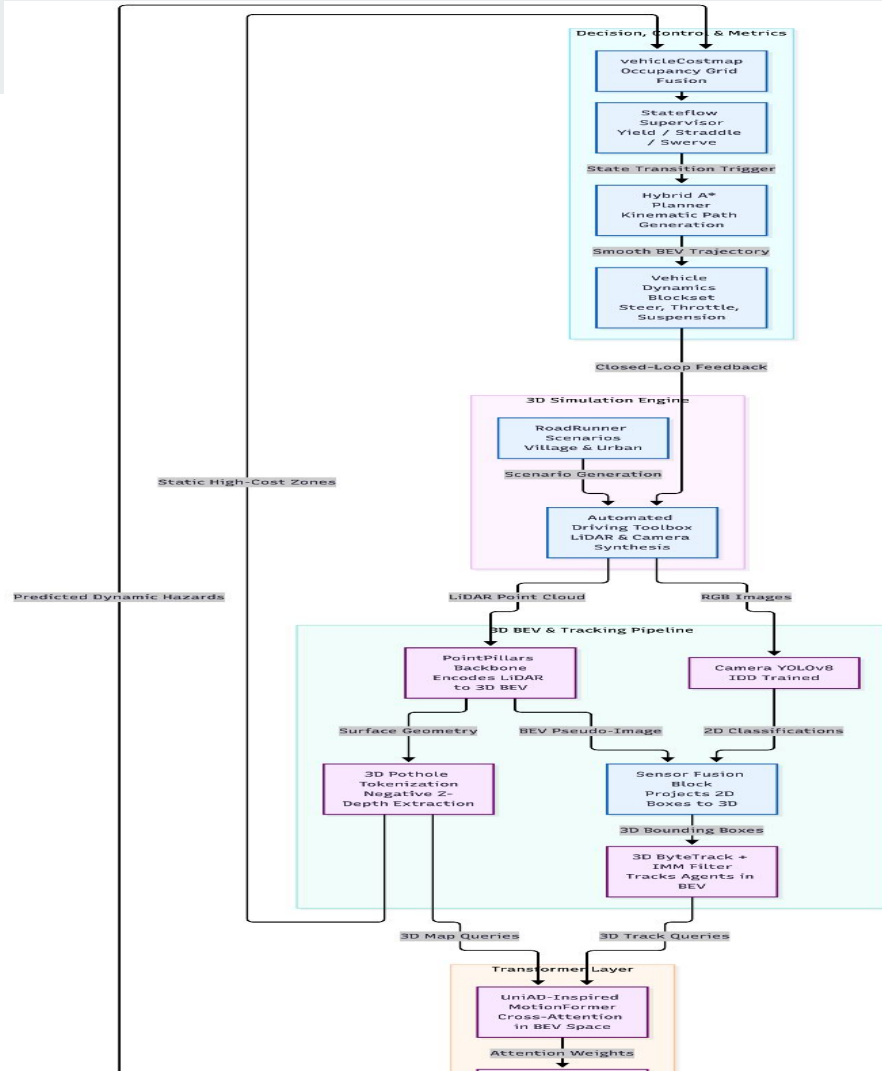
DEPTH ERROR

Achieved with 205 pts/m² sparse clouds ($p=0.8563$), proving survey-grade accuracy with budget sensors.

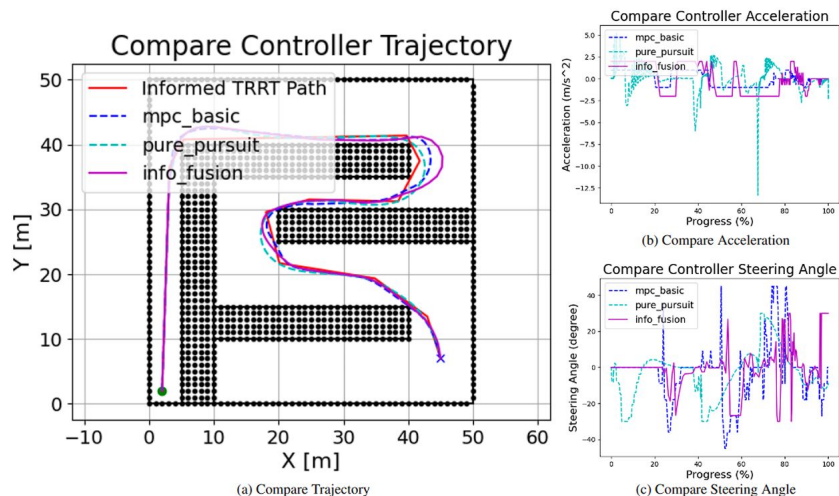


Path Planning/Decison





Vehicle Dynamics/control



Brief:

A planner can give a perfect path, but if the car cannot follow it smoothly, none of it matters. This track covers how the vehicle turns a planned path into real steering, throttle and brake commands, and how it stays stable even when that path changes suddenly.

Challenges:

- Sudden replans, like a cattle crossing or emergency stop, can jerk the steering and brakes instantly. This happens way too often in indian roads.
- Instant jerks feel unsafe and uncomfortable for passengers
- Path smoothness and completion rate are both scored by the PS, and both depend on how well the controller executes the plan
- A good plan can still fail in execution if the controller cannot track it accurately

Reference Image: Pure Pursuit stays flat for most of the run but has one sharp acceleration spike entering a tight turn. MPC avoids that single spike but oscillates continuously in both acceleration and steering throughout the run, meaning frequent, unpredictable jerks rather than one bad moment.

Proposed Approach: Adaptive Speed-Based Control

Plan:

- Pure Pursuit as default controller for slow, cluttered conditions, village roads, market areas, tight turns
- MPC takes over at higher speed, mainly highway merge, where Pure Pursuit is known to get unstable

Solution:

- Switch decided by vehicle speed, not by scenario, so it adapts automatically as conditions change
- A short blend added at the switch point so the handover itself does not cause a jerk
- Pure Pursuit upgraded to A-PP, lookahead distance adjusts on its own based on speed, curvature and lateral error, instead of staying fixed
- A-PP cuts lateral error notably over standard Pure Pursuit, with no added computation cost (as noted in reference images)

Table 3. Experiment error comparison table.

Method	Average Value of Lateral Error (m)	Variance (m)	Maximum Lateral Error (m)
Pure pursuit	0.02372	0.016588	0.036705
MPC	0.02069	0.014416	0.031628
A-PP	0.01070	0.006663	0.019443



THANK YOU!!